Chapter 7
Blocking and Confounding in the 2^k Factorial Design
Solutions

7.1S. Consider the experiment described in Problem 6.5. Analyze this experiment assuming that each replicate represents a block of a single production shift.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Cutting Speed (*A*)	0.67	1	0.67	<1
Tool Geometry (*B*)	770.67	1	770.67	22.38*
Cutting Angle (*C*)	280.17	1	280.17	8.14*
AB	16.67	1	16.67	<1
AC	468.17	1	468.17	13.60*
BC	48.17	1	48.17	1.40
ABC	28.17	1	28.17	<1
Blocks	0.58	2	0.29	
Error	482.08	14	34.43	
Total	2095.33	23		

Design Expert Output

Response: Life in hours

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Block	0.58	2	0.29			
Model	1519.67	4	379.92	11.23	0.0001	significant
A	*0.67*	*1*	*0.67*	*0.020*	*0.8900*	
B	*770.67*	*1*	*770.67*	*22.78*	*0.0002*	
C	*280.17*	*1*	*280.17*	*8.28*	*0.0104*	
AC	*468.17*	*1*	*468.17*	*13.84*	*0.0017*	
Residual	575.08	17	33.83			
Cor Total	2095.33	23				

The Model F-value of 11.23 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case B, C, AC are significant model terms.

These results agree with the results from Problem 6.5. Tool geometry, cutting angle and the interaction between cutting speed and cutting angle are significant at the 5% level. The Design Expert program also includes factor *A*, cutting speed, in the model to preserve hierarchy.

7.2. Consider the experiment described in Problem 6.9. Analyze this experiment assuming that each one of the four replicates represents a block.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Bit Size (*A*)	1107.23	1	1107.23	364.22*
Cutting Speed (*B*)	227.26	1	227.26	74.76*
AB	303.63	1	303.63	99.88*
Blocks	44.36	3	14.79	
Error	27.36	9	3.04	
Total	1709.83	15		

These results agree with those from Problem 6.9. Bit size, cutting speed and their interaction are significant at the 1% level.

Design Expert Output

Response: Vibration

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Block	44.36	3	14.79			
Model	1638.11	3	546.04	179.61	< 0.0001	significant
A	*1107.23*	*1*	*1107.23*	*364.21*	*< 0.0001*	
B	*227.26*	*1*	*227.26*	*74.75*	*< 0.0001*	
AB	*303.63*	*1*	*303.63*	*99.88*	*< 0.0001*	
Residual	27.36	9	3.04			
Cor Total	1709.83	15				

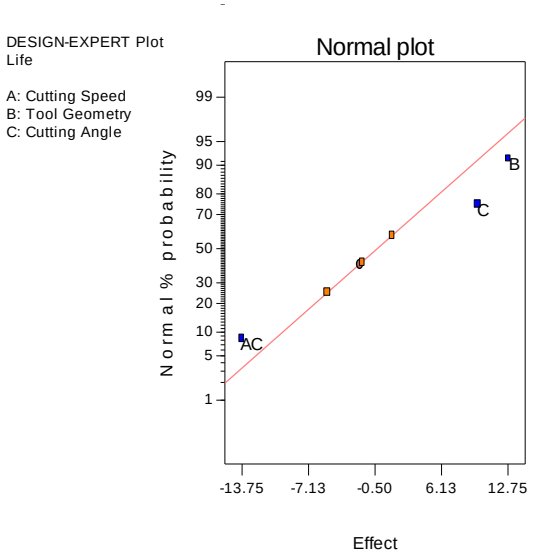
The Model F-value of 179.61 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, B, AB are significant model terms.

7.3S. Consider the data from the first replicate of Problem 6.5. Suppose that these observations could not all be run using the same bar stock. Set up a design to run these observations in two blocks of four observations each with *ABC* confounded. Analyze the data.

Block 1	Block 2
(1)	*a*
ab	*b*
ac	*c*
bc	*abc*

From the normal probability plot of effects, *B*, *C*, and the *AC* interaction are significant. Factor *A* was included in the analysis of variance to preserve hierarchy.



Design Expert Output

Response: Life in hours

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Block	91.13	1	91.13			
Model	896.50	4	224.13	7.32	0.1238	not significant
A	*3.13*	*1*	*3.13*	*0.10*	*0.7797*	
B	*325.12*	*1*	*325.12*	*10.62*	*0.0827*	
C	*190.12*	*1*	*190.12*	*6.21*	*0.1303*	
AC	*378.13*	*1*	*378.13*	*12.35*	*0.0723*	
Residual	61.25	2	30.62			
Cor Total	1048.88	7				

The "Model F-value" of 7.32 implies the model is not significant relative to the noise. There is a 12.38 % chance that a "Model F-value" this large could occur due to noise.

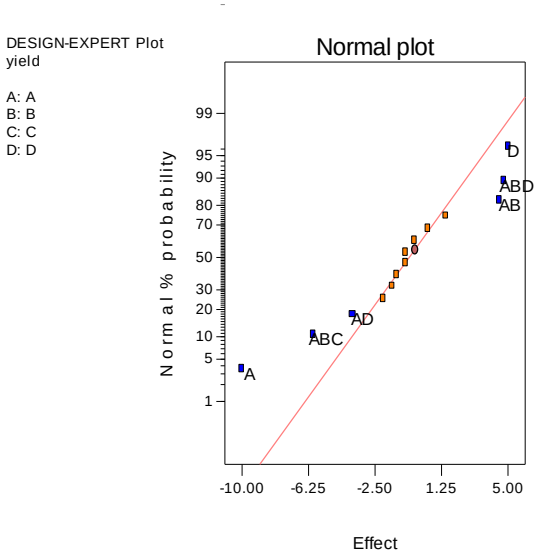
Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case there are no significant model terms.

This design identifies the same significant factors as Problem 6.5.

7.4. Consider the data from the first replicate of Problem 6.11. Construct a design with two blocks of eight observations each with *ABCD* confounded. Analyze the data.

Block 1	Block 2
(1)	*a*
ab	*b*
ac	*c*
bc	*d*
ad	*abc*
bd	*abd*
cd	*acd*
abcd	*bcd*

The significant effects are identified in the normal probability plot of effects below:



AC, *BC*, and *BD* were included in the model to preserve hierarchy.

Design Expert Output

Response: yield
ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Block	42.25	1	42.25			
Model	892.25	11	81.11	9.64	0.0438	significant
A	*400.00*	*1*	*400.00*	*47.52*	*0.0063*	
B	*2.25*	*1*	*2.25*	*0.27*	*0.6408*	
C	*2.25*	*1*	*2.25*	*0.27*	*0.6408*	
D	*100.00*	*1*	*100.00*	*11.88*	*0.0410*	
AB	*81.00*	*1*	*81.00*	*9.62*	*0.0532*	
AC	*1.00*	*1*	*1.00*	*0.12*	*0.7531*	
AD	*56.25*	*1*	*56.25*	*6.68*	*0.0814*	
BC	*6.25*	*1*	*6.25*	*0.74*	*0.4522*	
BD	*9.00*	*1*	*9.00*	*1.07*	*0.3772*	
ABC	*144.00*	*1*	*144.00*	*17.11*	*0.0256*	
ABD	*90.25*	*1*	*90.25*	*10.72*	*0.0466*	
Residual	25.25	3	8.42			
Cor Total	959.75	15				

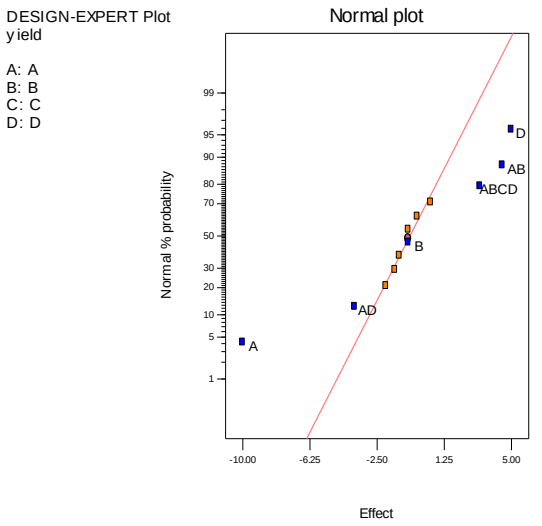
The Model F-value of 9.64 implies the model is significant. There is only a 4.38% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, D, ABC, ABD are significant model terms.

7.5. Repeat Problem 7.4 assuming that four blocks are required. Confound *ABD* and *ABC* (and consequently *CD*) with blocks.

The block assignments are shown in the table below. The normal probability plot of effects identifies factors *A* and *D*, and the interactions *AB*, *AD*, and the *ABCD* as strong candidates for the model. For hierarchal purposes, factor *B* was included in the model; however, hierarchy is not preserved for the *ABCD* interaction allowing an estimate for error.

Block 1	Block 2	Block 3	Block 4
(1)	*ac*	*c*	*a*
ab	*bc*	*abc*	*b*
acd	*d*	*ad*	*cd*
bcd	*abd*	*bd*	*abcd*



Design Expert Output

Response: **yield**

ANOVA for Selected Factorial Model
Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Block	243.25	3	81.08			
Model	681.75	6	113.63	19.62	0.0011	significant
A	*400.00*	*1*	*400.00*	*69.06*	*0.0002*	
B	*2.25*	*1*	*2.25*	*0.39*	*0.5560*	
D	*100.00*	*1*	*100.00*	*17.27*	*0.0060*	
AB	*81.00*	*1*	*81.00*	*13.99*	*0.0096*	
AD	*56.25*	*1*	*56.25*	*9.71*	*0.0207*	
ABCD	*42.25*	*1*	*42.25*	*7.29*	*0.0355*	
Residual	34.75	6	5.79			
Cor Total	959.75	15				

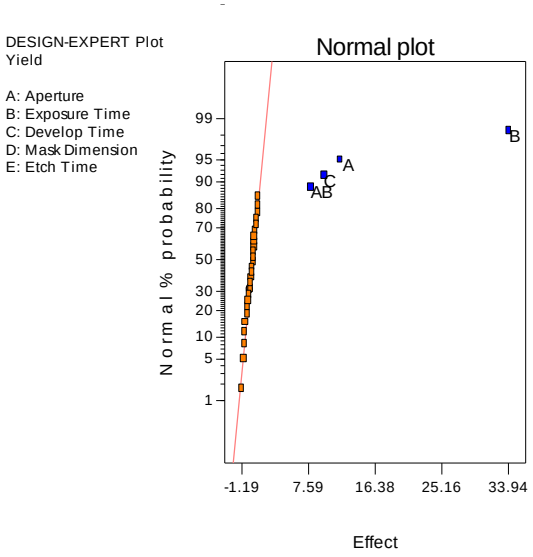
The Model F-value of 19.62 implies the model is significant. There is only a 0.11% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, D, AB, AD, ABCD are significant model terms.

7.6S. Using the data from the 2^5 design in Problem 6.24, construct and analyze a design in two blocks with *ABCDE* confounded with blocks.

Block 1	Block 1	Block 2	Block 2
(1)	*ae*	*a*	*e*
ab	*be*	*b*	*abe*
ac	*ce*	*c*	*ace*
bc	*abce*	*abc*	*bce*
ad	*de*	*d*	*ade*
bd	*abde*	*abd*	*bde*
cd	*acde*	*acd*	*cde*
abcd	*bcde*	*bcd*	*abcde*

The normal probability plot of effects identifies factors *A*, *B*, *C*, and the *AB* interaction as being significant. This is confirmed with the analysis of variance.



Design Expert Output

Response: Yield
ANOVA for Selected Factorial Model
Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Block	0.28	1	0.28			
Model	11585.13	4	2896.28	958.51	< 0.0001	significant
A	*1116.28*	*1*	*1116.28*	*369.43*	*< 0.0001*	
B	*9214.03*	*1*	*9214.03*	*3049.35*	*< 0.0001*	
C	*750.78*	*1*	*750.78*	*248.47*	*< 0.0001*	
AB	*504.03*	*1*	*504.03*	*166.81*	*< 0.0001*	
Residual	78.56	26	3.02			
Cor Total	11663.97	31				

The Model F-value of 958.51 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

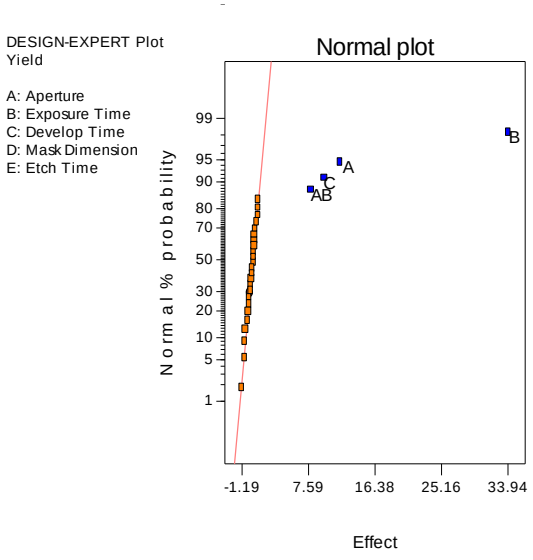
Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, B, C, AB are significant model terms.

7.7S. Repeat Problem 7.6S assuming that four blocks are necessary. Suggest a reasonable confounding scheme.

Use *ABC* and *CDE*, and consequently *ABDE*. The four blocks follow.

Block 1	Block 2	Block 3	Block 4
(1)	*a*	*ac*	*c*
ab	*b*	*bc*	*abc*
acd	*cd*	*d*	*ad*
bcd	*abcd*	*abd*	*bd*
ace	*ce*	*e*	*ae*
bce	*abce*	*abe*	*be*
de	*ade*	*acde*	*cde*
abde	*bde*	*bcde*	*abcde*

The normal probability plot of effects identifies the same significant effects as in Problem 7.7.



Design Expert Output

Response: Yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F

Block	13.84	3	4.61			
Model	11585.13	4	2896.28	1069.40	< 0.0001	significant
A	*1116.28*	*1*	*1116.28*	*412.17*	*< 0.0001*	
B	*9214.03*	*1*	*9214.03*	*3402.10*	*< 0.0001*	
C	*750.78*	*1*	*750.78*	*277.21*	*< 0.0001*	
AB	*504.03*	*1*	*504.03*	*186.10*	*< 0.0001*	
Residual	65.00	24	2.71			
Cor Total	11663.97	31				

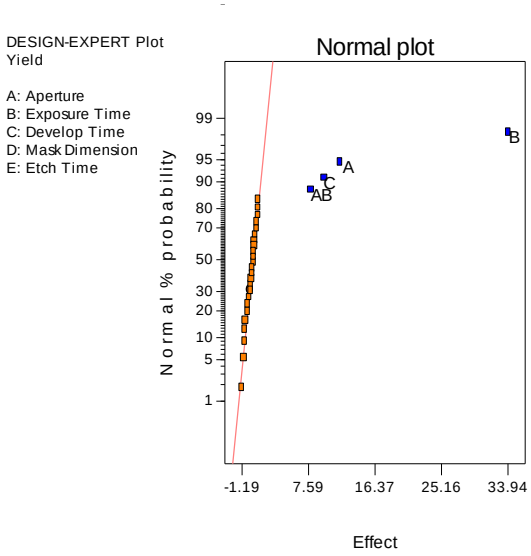
The Model F-value of 1069.40 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, B, C, AB are significant model terms.

7.8S. Consider the data from the 2^5 design in Problem 6.24. Suppose that it was necessary to run this design in four blocks with *ACDE* and *BCD* (and consequently *ABE*) confounded. Analyze the data from this design.

Block 1	Block 2	Block 3	Block 4
(1)	*a*	*b*	*c*
ae	*e*	*abe*	*ace*
cd	*acd*	*bcd*	*d*
abc	*bc*	*ac*	*ab*
acde	*cde*	*abcde*	*ade*
bce	*abce*	*ce*	*be*
abd	*bd*	*ad*	*abcd*
bde	*abde*	*de*	*bcde*

Even with four blocks, the same effects are identified as significant per the normal probability plot and analysis of variance below:



Design Expert Output

Response: Yield

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Block	2.59	3	0.86			
Model	11585.13	4	2896.28	911.62	< 0.0001	significant
A	*1116.28*	*1*	*1116.28*	*351.35*	*< 0.0001*	

B	*9214.03*	*1*	*9214.03*	*2900.15*	*< 0.0001*
C	*750.78*	*1*	*750.78*	*236.31*	*< 0.0001*
AB	*504.03*	*1*	*504.03*	*158.65*	*< 0.0001*
Residual	76.25	24	3.18		
Cor Total	11663.97	31			

The Model F-value of 911.62 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, B, C, AB are significant model terms.

7.9S. Consider the fill height deviation experiment in Problem 6.20. Suppose that each replicate was run on a separate day. Analyze the data assuming that the days are blocks.

Design Expert Output

Response: Fill Deviation

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Block	1.00	1	1.00			
Model	70.75	4	17.69	28.30	< 0.0001	significant
A	*36.00*	*1*	*36.00*	*57.60*	*< 0.0001*	
B	*20.25*	*1*	*20.25*	*32.40*	*0.0002*	
C	*12.25*	*1*	*12.25*	*19.60*	*0.0013*	
AB	*2.25*	*1*	*2.25*	*3.60*	*0.0870*	
Residual	6.25	10	0.62			
Cor Total	78.00	15				

The Model F-value of 28.30 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, B, C are significant model terms.

The analysis is very similar to the original analysis in chapter 6. The same effects are significant.

7.10. Consider the fill height deviation experiment in Problem 6.20. Suppose that only four runs could be made on each shift. Set up a design with *ABC* confounded in replicate 1 and *AC* confounded in replicate 2. Analyze the data and comment on your findings.

Design Expert Output

Response: Fill Deviation

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Block	1.50	3	0.50			
Model	70.75	4	17.69	24.61	0.0001	significant
A	*36.00*	*1*	*36.00*	*50.09*	*0.0001*	
B	*20.25*	*1*	*20.25*	*28.17*	*0.0007*	
C	*12.25*	*1*	*12.25*	*17.04*	*0.0033*	
AB	*2.25*	*1*	*2.25*	*3.13*	*0.1148*	
Residual	5.75	8	0.72			
Cor Total	78.00	15				

The Model F-value of 24.61 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, B, C are significant model terms.

The analysis is very similar to the original analysis of Problem 6.20 and that of problem 7.9S. The *AB* interaction is less significant in this scenario.

7.11S. Consider the putting experiment in Problem 6.21. Analyze the data considering each replicate as a block.

The analysis is similar to that of Problem 6.21. Blocking has not changed the significant factors, however, the residual plots show that the normality assumption has been violated. The transformed data also has similar analysis to the transformed data of Problem 6.21. The ANOVA shown is for the transformed data.

Design Expert Output

Response: Distance from cupTransform:Square root Constant: 0

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	Sum of Squares	DF	Mean Square	F Value	Prob > F	
Block	13.50	6	2.25			
Model	37.26	2	18.63	7.83	0.0007	significant
A	21.61	1	21.61	9.08	0.0033	
B	15.64	1	15.64	6.57	0.0118	
Residual	245.13	103	2.38			
Cor Total	295.89	111				

The Model F-value of 7.83 implies the model is significant. There is only a 0.07% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, B are significant model terms.

7.12. The experiment in Problem 6.35 is a 2^5 factorial. Suppose that this design had been run in four blocks of eight runs each.

(a) Recommend a blocking scheme and set up the design.

Interactions *ABC* and *BDE* are confounded with the blocks such that:

Block	*ABC*	*BDE*
1	–	+
2	+	–
3	–	–
4	+	+

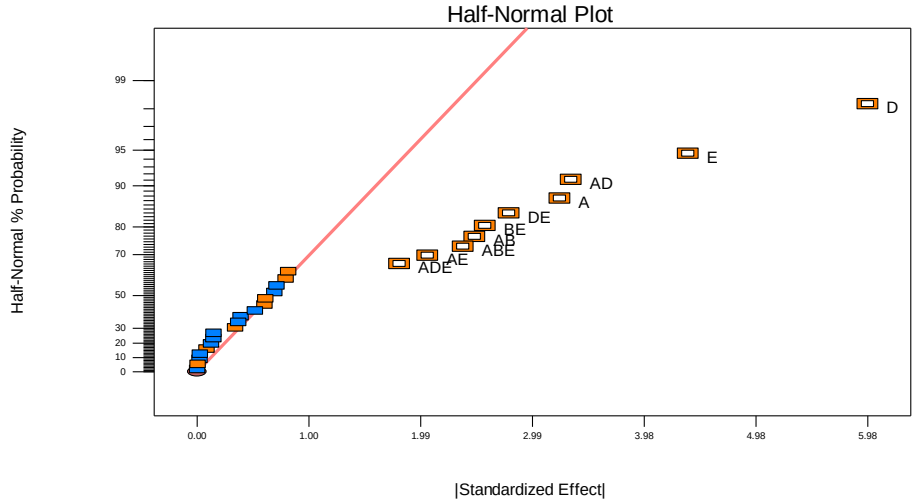
Note, the *ACDE* interaction is also confounded with the blocks. The experimental runs with the blocks are shown below.

Block	A	B	C	D	E	y
Block 1	-1	-1	-1	-1	-1	8.11
Block 2	1	-1	-1	-1	-1	5.56
Block 4	-1	1	-1	-1	-1	5.77
Block 3	1	1	-1	-1	-1	5.82
Block 2	-1	-1	1	-1	-1	9.17
Block 1	1	-1	1	-1	-1	7.8
Block 3	-1	1	1	-1	-1	3.23
Block 4	1	1	1	-1	-1	5.69
Block 3	-1	-1	-1	1	-1	8.82
Block 4	1	-1	-1	1	-1	14.23
Block 2	-1	1	-1	1	-1	9.2
Block 1	1	1	-1	1	-1	8.94
Block 4	-1	-1	1	1	-1	8.68

Block 3	1	-1	1	1	-1	11.49
Block 1	-1	1	1	1	-1	6.25
Block 2	1	1	1	1	-1	9.12
Block 3	-1	-1	-1	-1	1	7.93
Block 4	1	-1	-1	-1	1	5
Block 2	-1	1	-1	-1	1	7.47
Block 1	1	1	-1	-1	1	12
Block 4	-1	-1	1	-1	1	9.86
Block 3	1	-1	1	-1	1	3.65
Block 1	-1	1	1	-1	1	6.4
Block 2	1	1	1	-1	1	11.61
Block 1	-1	-1	-1	1	1	12.43
Block 2	1	-1	-1	1	1	17.55
Block 4	-1	1	-1	1	1	8.87
Block 3	1	1	-1	1	1	25.38
Block 2	-1	-1	1	1	1	13.06
Block 1	1	-1	1	1	1	18.85
Block 3	-1	1	1	1	1	11.78
Block 4	1	1	1	1	1	26.05

(b) Analyze the data from this blocked design. Is blocking important?

Blocking does not appear to be important; however, if the *ADE* or *ABE* interaction had been chosen to define the blocks, then blocking would have appeared as important. The *ADE* and *ABE* are significant effects in the analysis below.



Design Expert Output

Response 1 y

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	2.58	3	0.86			
Model	879.62	11	79.97	45.38	< 0.0001	significant
A-A	*83.56*	*1*	*83.56*	*47.41*	*< 0.0001*	
B-B	*0.060*	*1*	*0.060*	*0.034*	*0.8553*	
D-D	*285.78*	*1*	*285.78*	*162.16*	*< 0.0001*	
E-E	*153.17*	*1*	*153.17*	*86.91*	*< 0.0001*	
AB	*48.93*	*1*	*48.93*	*27.76*	*< 0.0001*	
AD	*88.88*	*1*	*88.88*	*50.43*	*< 0.0001*	
AE	*33.76*	*1*	*33.76*	*19.16*	*0.0004*	
BE	*52.71*	*1*	*52.71*	*29.91*	*< 0.0001*	
DE	*61.80*	*1*	*61.80*	*35.07*	*< 0.0001*	

ABE	*44.96*	*1*	*44.96*	*25.51*	*< 0.0001*
ADE	*26.01*	*1*	*26.01*	*14.76*	*0.0013*
Residual	29.96	17	1.76		
Cor Total	912.16	31			

The Model F-value of 45.38 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, D, E, AB, AD, AE, BE, DE, ABE, ADE are significant model terms.

7.13. Repeat Problem 7.12 using a design in two blocks.

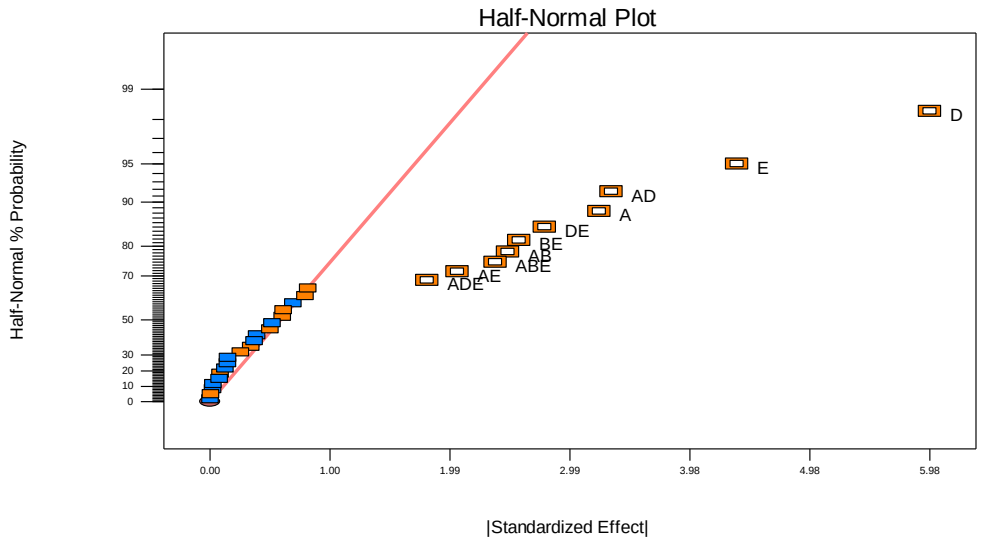
(a) Recommend a blocking scheme and set up the design.

Interaction *ABCDE* is confounded with the blocks. The design is shown below.

Block	A	B	C	D	E	y
Block 1	-1	-1	-1	-1	-1	8.11
Block 2	1	-1	-1	-1	-1	5.56
Block 2	-1	1	-1	-1	-1	5.77
Block 1	1	1	-1	-1	-1	5.82
Block 2	-1	-1	1	-1	-1	9.17
Block 1	1	-1	1	-1	-1	7.8
Block 1	-1	1	1	-1	-1	3.23
Block 2	1	1	1	-1	-1	5.69
Block 2	-1	-1	-1	1	-1	8.82
Block 1	1	-1	-1	1	-1	14.23
Block 1	-1	1	-1	1	-1	9.2
Block 2	1	1	-1	1	-1	8.94
Block 1	-1	-1	1	1	-1	8.68
Block 2	1	-1	1	1	-1	11.49
Block 2	-1	1	1	1	-1	6.25
Block 1	1	1	1	1	-1	9.12
Block 2	-1	-1	-1	-1	1	7.93
Block 1	1	-1	-1	-1	1	5
Block 1	-1	1	-1	-1	1	7.47
Block 2	1	1	-1	-1	1	12
Block 1	-1	-1	1	-1	1	9.86
Block 2	1	-1	1	-1	1	3.65
Block 2	-1	1	1	-1	1	6.4
Block 1	1	1	1	-1	1	11.61
Block 1	-1	-1	-1	1	1	12.43
Block 2	1	-1	-1	1	1	17.55
Block 2	-1	1	-1	1	1	8.87
Block 1	1	1	-1	1	1	25.38
Block 2	-1	-1	1	1	1	13.06
Block 1	1	-1	1	1	1	18.85
Block 1	-1	1	1	1	1	11.78
Block 2	1	1	1	1	1	26.05

(b) Analyze the data from this blocked design. Is blocking important?

The analysis below shows that the blocking does not appear to be very important.



Design Expert Output

Response 1 y

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	4.04	1	4.04			
Model	879.62	11	79.97	53.31	< 0.0001	significant
A-A	*83.56*	*1*	*83.56*	*55.71*	*< 0.0001*	
B-B	*0.060*	*1*	*0.060*	*0.040*	*0.8431*	
D-D	*285.78*	*1*	*285.78*	*190.54*	*< 0.0001*	
E-E	*153.17*	*1*	*153.17*	*102.12*	*< 0.0001*	
AB	*48.93*	*1*	*48.93*	*32.62*	*< 0.0001*	
AD	*88.88*	*1*	*88.88*	*59.26*	*< 0.0001*	
AE	*33.76*	*1*	*33.76*	*22.51*	*0.0001*	
BE	*52.71*	*1*	*52.71*	*35.14*	*< 0.0001*	
DE	*61.80*	*1*	*61.80*	*41.20*	*< 0.0001*	
ABE	*44.96*	*1*	*44.96*	*29.98*	*< 0.0001*	
ADE	*26.01*	*1*	*26.01*	*17.34*	*0.0005*	
Residual	28.50	19	1.50			
Cor Total	912.16	31				

The Model F-value of 53.31 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant.
In this case A, D, E, AB, AD, AE, BE, DE, ABE, ADE are significant model terms.

7.14. The design in Problem 6.38 is a 2^3 factorial replicated twice. Suppose that each replicate was a block. Analyze all of the responses from this blocked design. Are the results comparable to those from Problem 6.46? Is the block effect large?

The block effect is not large and does not appear to be important for the analysis on any of the four the responses as shown below. The results are comparable to those from Problem 6.46.

Design Expert Output

Response 1 Fishbone Pb

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	0.000	1	0.000			
Model	12.19	7	1.74	2300.73	< 0.0001	significant
A-Apatite	*10.99*	*1*	*10.99*	*14514.07*	*< 0.0001*	
B-pH	*0.35*	*1*	*0.35*	*459.75*	*< 0.0001*	
C-Pb	*0.27*	*1*	*0.27*	*350.30*	*< 0.0001*	
AB	*0.36*	*1*	*0.36*	*475.47*	*< 0.0001*	
AC	*0.19*	*1*	*0.19*	*249.92*	*< 0.0001*	
BC	*0.022*	*1*	*0.022*	*29.72*	*0.0010*	
ABC	*0.020*	*1*	*0.020*	*25.89*	*0.0014*	
Residual	5.300E-003	7	7.571E-004			
Cor Total	12.20	15				

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	0.85	1	6.879E-003	0.84	0.87	
Block 1	0.000	1				
Block 2	0.000					
A-Apatite	-0.83	1	6.879E-003	-0.85	-0.81	1.00
B-pH	-0.15	1	6.879E-003	-0.16	-0.13	1.00
C-Pb	0.13	1	6.879E-003	0.11	0.15	1.00
AB	0.15	1	6.879E-003	0.13	0.17	1.00
AC	-0.11	1	6.879E-003	-0.13	-0.092	1.00
BC	0.037	1	6.879E-003	0.021	0.054	1.00
ABC	-0.035	1	6.879E-003	-0.051	-0.019	1.00

Design Expert Output

Response 1 Fishbone pH

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	2.256E-003	1	2.256E-003			
Model	21.09	7	3.01	102.87	< 0.0001	significant
A-Apatite	*9.84*	*1*	*9.84*	*336.14*	*< 0.0001*	
B-pH	*8.14*	*1*	*8.14*	*277.85*	*< 0.0001*	
C-Pb	*1.12*	*1*	*1.12*	*38.19*	*0.0005*	
AB	*0.61*	*1*	*0.61*	*20.91*	*0.0026*	
AC	*1.17*	*1*	*1.17*	*40.01*	*0.0004*	
BC	*0.098*	*1*	*0.098*	*3.33*	*0.1106*	
ABC	*0.11*	*1*	*0.11*	*3.66*	*0.0972*	
Residual	0.20	7	0.029			
Cor Total	21.30	15				

The Model F-value of 102.87 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, B, C, AB, AC are significant model terms.

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	5.05	1	0.043	4.95	5.15	
Block 1	-0.012	1				
Block 2	0.012					
A-Apatite	0.78	1	0.043	0.68	0.89	1.00
B-pH	-0.71	1	0.043	-0.81	-0.61	1.00
C-Pb	-0.26	1	0.043	-0.37	-0.16	1.00
AB	0.20	1	0.043	0.094	0.30	1.00
AC	-0.27	1	0.043	-0.37	-0.17	1.00
BC	-0.078	1	0.043	-0.18	0.023	1.00
ABC	-0.082	1	0.043	-0.18	0.019	1.00

Design Expert Output

Response 1 Hydroxyapatite Pb

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	2.250E-004	1	2.250E-004			
Model	4.01	7	0.57	937.82	< 0.0001	significant
A-Apatite	*2.45*	*1*	*2.45*	*4010.43*	*< 0.0001*	
B-pH	*0.27*	*1*	*0.27*	*434.29*	*< 0.0001*	
C-Pb	*0.54*	*1*	*0.54*	*884.58*	*< 0.0001*	
AB	*0.17*	*1*	*0.17*	*275.25*	*< 0.0001*	
AC	*0.50*	*1*	*0.50*	*825.43*	*< 0.0001*	
BC	*0.036*	*1*	*0.036*	*59.11*	*0.0001*	
ABC	*0.046*	*1*	*0.046*	*75.69*	*< 0.0001*	
Residual	4.275E-003	7	6.107E-004			
Cor Total	4.01	15				

The Model F-value of 937.82 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, B, C, AB, AC, BC, ABC are significant model terms.

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	0.42	1	6.178E-003	0.40	0.43	
Block 1	3.750E-003	1				
Block 2	-3.750E-003					
A-Apatite	-0.39	1	6.178E-003	-0.41	-0.38	1.00
B-pH	0.13	1	6.178E-003	0.11	0.14	1.00
C-Pb	0.18	1	6.178E-003	0.17	0.20	1.00
AB	-0.10	1	6.178E-003	-0.12	-0.088	1.00
AC	-0.18	1	6.178E-003	-0.19	-0.16	1.00
BC	-0.048	1	6.178E-003	-0.062	-0.033	1.00
ABC	0.054	1	6.178E-003	0.039	0.068	1.00

Design Expert Output

Response 1 Hydroxyapatite pH

ANOVA for selected factorial model

Analysis of variance table [Partial sum of squares - Type III]

Source	Sum of Squares	df	Mean Square	F Value	p-value Prob > F	
Block	2.025E-003	1	2.025E-003			
Model	20.44	7	2.92	1494.46	< 0.0001	significant
A-Apatite	*8.15*	*1*	*8.15*	*4172.37*	*< 0.0001*	
B-pH	*8.82*	*1*	*8.82*	*4515.27*	*< 0.0001*	
C-Pb	*0.084*	*1*	*0.084*	*43.05*	*0.0003*	
AB	*3.24*	*1*	*3.24*	*1658.50*	*< 0.0001*	
AC	*0.014*	*1*	*0.014*	*7.37*	*0.0300*	
BC	*0.13*	*1*	*0.13*	*64.51*	*< 0.0001*	
ABC	*2.250E-004*	*1*	*2.250E-004*	*0.12*	*0.7443*	
Residual	0.014	7	1.954E-003			
Cor Total	20.45	15				

Factor	Coefficient Estimate	df	Standard Error	95% CI Low	95% CI High	VIF
Intercept	3.77	1	0.011	3.74	3.79	
Block 1	0.011	1				
Block 2	-0.011					
A-Apatite	0.71	1	0.011	0.69	0.74	1.00
B-pH	-0.74	1	0.011	-0.77	-0.72	1.00
C-Pb	-0.073	1	0.011	-0.099	-0.046	1.00
AB	-0.45	1	0.011	-0.48	-0.42	1.00
AC	-0.030	1	0.011	-0.056	-3.871E-003	1.00
BC	0.089	1	0.011	0.063	0.11	1.00
ABC	3.750E-003	1	0.011	-0.022	0.030	1.00

7.15. Consider the 2^6 design in eight blocks of eight runs each with *ABCD*, *ACE*, and *ABEF* as the independent effects chosen to be confounded with blocks. Generate the design. Find the other effects confound with blocks.

Block 1	Block 2	Block 3	Block 4	Block 5	Block 6	Block 7	Block 8
b	*abc*	*a*	*c*	*ac*	(1)	*bc*	*ab*
acd	*d*	*bcd*	*abd*	*bd*	*abcd*	*ad*	*cd*
ce	*ae*	*abce*	*be*	*abe*	*bce*	*e*	*ace*
abde	*bcde*	*de*	*acde*	*cde*	*ade*	*abcde*	*bde*
abcf	*bf*	*cf*	*af*	*f*	*acf*	*abf*	*bcf*
df	*acdf*	*abdf*	*bcdf*	*abcdf*	*bdf*	*cdf*	*adf*
aef	*cef*	*bef*	*abcef*	*bcef*	*abef*	*acef*	*ef*
bcdef	*abdef*	*acdef*	*def*	*adef*	*cdef*	*bdef*	*abcdef*

The factors that are confounded with blocks are *ABCD*, *ABEF*, *ACE*, *BDE*, *CDEF*, *BCF*, and *ADF*.

7.16S. Consider the 2^2 design in two blocks with *AB* confounded. Prove algebraically that $SS_{AB} = SS_{\text{Blocks}}$. If *AB* is confounded, the two blocks are:

Block 1	Block 2
(1)	*a*
ab	*b*
(1) + *ab*	*a* + *b*

$$SS_{Blocks} = \frac{\left[(1)+ab\right]^2+\left[a+b\right]^2}{2} - \frac{\left[(1)+ab+a+b\right]^2}{4}$$

$$SS_{Blocks} = \frac{(1)^2+(ab)^2+2(1)ab+a^2+b^2+2ab}{2} - \frac{(1)^2+(ab)^2+a^2+b^2+2(1)ab+2(1)a+2(1)b+2a(ab)+2b(ab)+2ab}{4}$$

$$SS_{Blocks} = \frac{(1)^2+(ab)^2+a^2+b^2+2(1)ab+2ab-2(1)a-2(1)b-2a(ab)-2b(ab)}{4}$$

$$SS_{Blocks} = \frac{1}{4}\left[(1)+ab-a-b\right]^2 = SS_{AB}$$

7.17. Consider the data in Example 7.2. Suppose that all the observations in block 2 are increased by 20. Analyze the data that would result. Estimate the block effect. Can you explain its magnitude? Do blocks now appear to be an important factor? Are any other effect estimates impacted by the change you made in the data?

$$\text{Block Effect} = \bar{y}_{Block\,1} - \bar{y}_{Block\,2} = \frac{406}{8} - \frac{715}{8} = \frac{-309}{8} = -38.625$$

This is the block effect estimated in Example 7.2 plus the additional 20 units that were added to each observation in block 2. All other effects are the same.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
A	1870.56	1	1870.56	89.93
C	390.06	1	390.06	18.75
D	855.56	1	855.56	41.13
AC	1314.06	1	1314.06	63.18
AD	1105.56	1	1105.56	53.15
Blocks	5967.56	1	5967.56	
Error	187.56	9	20.8	
Total	11690.9	15		

Design Expert Output

Response: Filtration in gal/hr

ANOVA for Selected Factorial Model

Analysis of variance table [Partial sum of squares]

Source	**Sum of Squares**	**DF**	**Mean Square**	**F Value**	**Prob > F**	
Block	5967.56	1	5967.56			
Model	5535.81	5	1107.16	53.13	< 0.0001	significant
A	*1870.56*	*1*	*1870.56*	*89.76*	*< 0.0001*	
C	*390.06*	*1*	*390.06*	*18.72*	*0.0019*	
D	*855.56*	*1*	*855.56*	*41.05*	*0.0001*	
AC	*1314.06*	*1*	*1314.06*	*63.05*	*< 0.0001*	
AD	*1105.56*	*1*	*1105.56*	*53.05*	*< 0.0001*	
Residual	187.56	9	20.84			
Cor Total	11690.94	15				

The Model F-value of 53.13 implies the model is significant. There is only a 0.01% chance that a "Model F-Value" this large could occur due to noise.

Values of "Prob > F" less than 0.0500 indicate model terms are significant. In this case A, C, D, AC, AD are significant model terms.

7.18. Suppose that the data in Problem 6.5 we had confounded *ABC* in replicate I, *AB* in replicate II, and *BC* in replicate III. Construct the analysis of variance table.

	Replicate I (*ABC* Confounded)		Replicate II (*AB* Confounded)		Replicate III (*BC* Confounded)	
Block->	1	2	1	2	1	2
	(1)	*a*	(1)	*a*	(1)	*b*
	ab	*b*	*ab*	*b*	*bc*	*c*
	ac	*c*	*abc*	*ac*	*abc*	*ab*
	bc	*abc*	*c*	*bc*	*a*	*ac*

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
A	0.67	1	0.67	<1
B	770.67	1	770.67	20.77
C	280.17	1	280.17	7.55
AB (reps 1 and III)	25.00	1	25.00	<1
AC	468.17	1	468.17	12.62
BC (reps I and II)	22.56	1	22.56	<1
ABC (reps II and III)	0.06	1	0.06	<1
Blocks within replicates	119.25	3	39.75	
Replicates	0.58	2		
Error	408.21	11	37.11	
Total	2095.33	23		

7.19. Repeat the analysis of Problem 6.5 assuming that *ABC* was confounded with blocks in each replicate.

	Replicate I, II, and III (*ABC* Confounded)	
Block->	1	2
	(1)	*a*
	ab	*b*
	ac	*c*
	bc	*abc*

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
A	0.67	1	0.67	<1
B	770.67	1	770.67	22.38
C	280.17	1	280.17	8.14
AB	16.67	1	16.67	<1
AC	468.17	1	468.17	13.59
BC	48.17	1	48.17	1.40
Blocks (or *ABC*)	28.17	1	28.17	
Replicates/Lack of Fit	0.58	2		
Error	482.09	14	34.44	
Total	2095.33	23		

7.20. Suppose that in Problem 6.11 *ABCD* was confounded in replicate I and *ABC* was confounded in replicate II. Perform the statistical analysis of variance.

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
A	657.03	1	657.03	84.89
B	13.78	1	13.78	1.78
C	57.78	1	57.78	7.46
D	124.03	1	124.03	16.02
AB	132.03	1	132.03	17.06
AC	3.78	1	3.78	<1
AD	38.28	1	38.28	4.95
BC	2.53	1	2.53	<1
BD	0.28	1	0.28	<1
CD	22.78	1	22.78	2.94
ABC	144.00	1	144.00	18.64
ABD	175.78	1	175.78	22.71
ACD	7.03	1	7.03	<1
BCD	7.03	1	7.03	<1
ABCD	10.56	1	10.56	1.36
Replicates	11.28	1	11.28	
Blocks	118.81	2	59.41	
Error	100.65	13	7.74	
Total	1627.47	31		

7.21. Construct a 2^3 design with *ABC* confounded in the first two replicates and *BC* confounded in the third. Outline the analysis of variance and comment on the information obtained.

	Replicate I (*ABC* Confounded)		Replicate II (*ABC* Confounded)		Replicate III (*BC* Confounded)	
Block->	1	2	1	2	1	2
	(1)	*a*	(1)	*a*	(1)	*b*
	ab	*b*	*ab*	*b*	*bc*	*c*
	ac	*c*	*ac*	*c*	*abc*	*ab*
	bc	*abc*	*bc*	*abc*	*a*	*ac*

Source of Variation	Degree Freedo
A	1
B	1
C	1
AB	1
AC	1
BC	1
ABC	1
Replicates	2
Blocks	3
Error	11
Total	23

This design provides "two-thirds" information on *BC* and "one-third" information on *ABC*.

7.22. The block effect in a two-level design with two blocks can be calculated directly as the difference in the average response between the two blocks.

True False

7.23. When constructing the 2^7 design confounded in 8 blocks, three independent effects are chosen to generate the blocks, and there are a total of 8 interaction confounded with blocks.

True **False**

7.24. Consider the 2^5 factor design in two blocks. If ABCDE is confounded with blocks, then which of the following runs is in the same block as run *acde*?

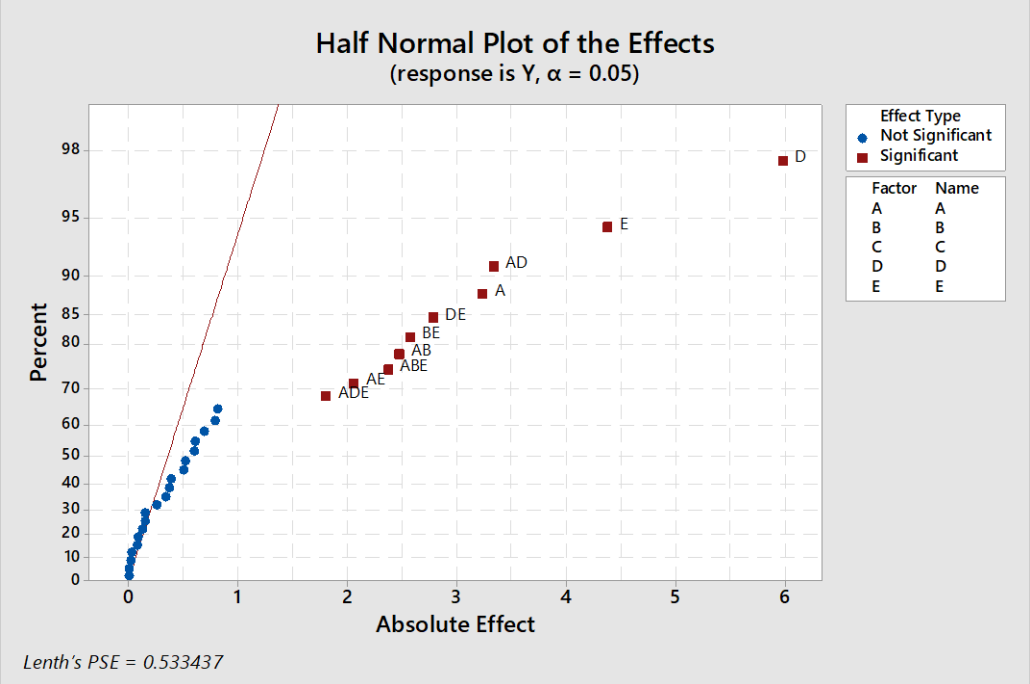
(a) *a*
(b) *acd*
(c) *bcd*
(d) *be*
(e) *abe*
(f) **None of the above**

7.25. The information on the interaction confounded with the block can always be separated from the block effect.

True **False**

7.26. Consider the full 2^5 factorial design problem in 6.42. Suppose that this experiment had been run in two blocks with *ABCDE* confounded with the blocks. Set up the blocked design and perform the analysis. Compare your results with the results obtained for the completely randomized design in Problem 6.42.

Minitab Output



Since no four factor interactions appear to be important, remove them from the model to get the ANOVA table. Also, Factor *C* and its interaction are not important and they are removed from the model.

Minitab Output

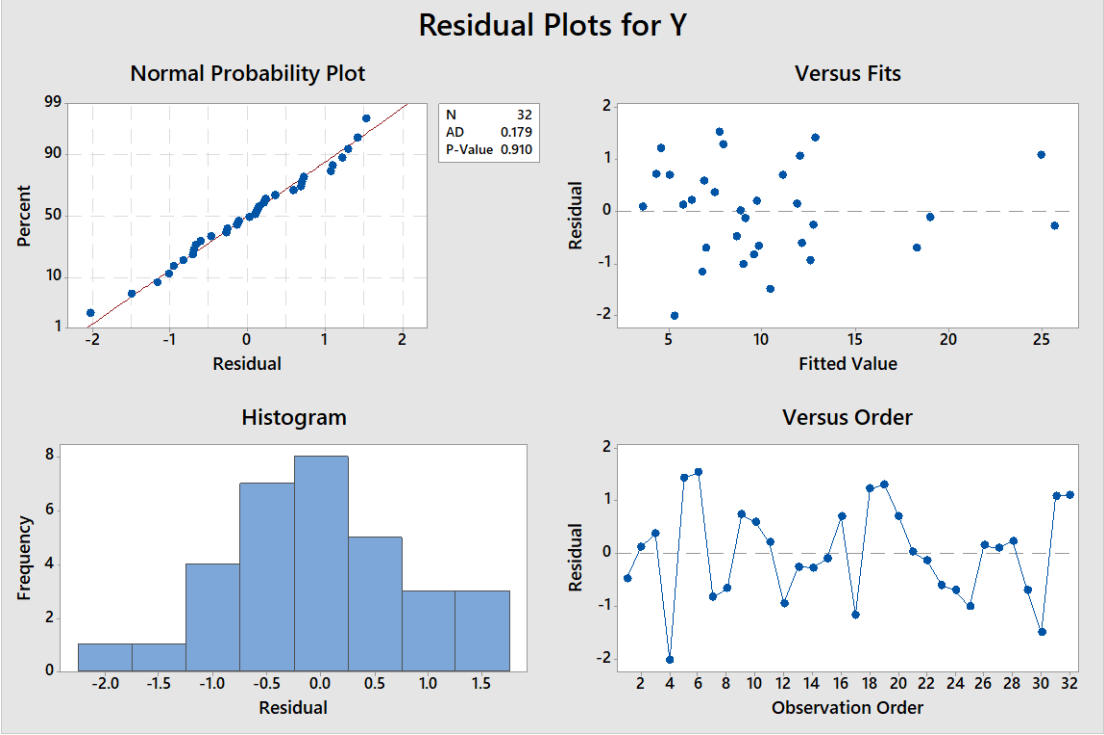
Factorial Regression: Y versus Blocks, A, B, D, E

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	15	887.537	59.169	38.44	0.000
Blocks	1	4.040	4.040	2.62	0.125
Linear	4	522.574	130.643	84.88	0.000
A	1	83.560	83.560	54.29	0.000
B	1	0.060	0.060	0.04	0.845
D	1	285.784	285.784	185.68	0.000
E	1	153.169	153.169	99.52	0.000
2-Way Interactions	6	286.088	47.681	30.98	0.000
A*B	1	48.931	48.931	31.79	0.000
A*D	1	88.878	88.878	57.75	0.000
A*E	1	33.764	33.764	21.94	0.000
B*D	1	0.006	0.006	0.00	0.952
B*E	1	52.711	52.711	34.25	0.000
D*E	1	61.799	61.799	40.15	0.000
3-Way Interactions	4	74.835	18.709	12.16	0.000
A*B*D	1	3.816	3.816	2.48	0.135
A*B*E	1	44.959	44.959	29.21	0.000
A*D*E	1	26.010	26.010	16.90	0.001
B*D*E	1	0.050	0.050	0.03	0.859
Error	16	24.626	1.539		
Total	31	912.162			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
1.24061	97.30%	94.77%	89.20%



The Block effect is not important and results are the same as Problem 6.42. The residuals are acceptable for this model.

7.27. Suppose that you are designing and experiment for four factors and that due to material properties it is necessary to conduct the experiment in blocks. Material availability restricts you to the use of two blocks; however, each batch of material is only sufficient for six runs. So, the standard 2^4 factorial in two blocks of eight runs each with *ABCD* confounded will not work. Recommend a design. Suggestion: this is a reasonable application for a *D*-optimal design. What type of design do you find in each block?

This is a reasonable application of the *D*-optimal Design. In the design below, the blocks are basically a fractional factorial with a couple of corner points missing.

JMP Table

Custom Design 2 - JMP Pro

File Edit Tables Rows Cols DOE Analyze Graph Tools View Window Help

Custom Design 2
Design Criterion: Custom Design
D Optimal
Model
Evaluate Design
DOE Dialog
Columns (6/0)

	Random Block	X1	X2	X3	X4	Y
1	1	-1	-1	1	-1	•
2	1	1	-1	-1	-1	•
3	1	1	1	1	1	•
4	1	1	-1	1	1	•
5	1	-1	1	-1	1	•
6	1	1	1	-1	-1	•
7	2	1	-1	1	-1	•
8	2	1	1	-1	1	•
9	2	-1	1	1	-1	•
10	2	-1	-1	-1	-1	•
11	2	-1	1	1	1	•
12	2	-1	-1	-1	1	•